State of the Art and Preliminary Design of an Autonomous Differential-Drive Ground Vehicle for the John Deere Challenge

Daniel Appel
Tecnológico de Monterrey
Mexico
A01833285

Thomas Michael Resch
Tecnológico de Monterrey
Mexico
A01832776

Jan Mosler
Tecnológico de Monterrey
Mexico
A01833812

Isaac Sebastian Hovmad
Tecnológico de Monterrey
Mexico
A01832781

***Abstract*—This paper documents the first design stage of an autonomous differential-drive ground vehicle for the John Deere challenge of course MR2006B. The battery-powered robot must complete an outdoor circuit of roughly 150 m, stop for 10 s at three waypoints, avoid obstacles and stay on paved paths, using an STM32 microcontroller as its only computer and no external positioning aids. The mechanical platform (two driven wheels, two passive casters), a geared brushed-DC drivetrain with Hall encoders and BTS7960 drivers, and a per-wheel cascade controller (trapezoidal profile, position loop, feedforward and PI speed loop) are implemented and have been identified on the bench. The proposed methodology fuses wheel odometry, a gyroscope and time-of-flight range sensors in an extended Kalman filter, and adds a cross-track path follower, a mission state machine and a radio telemetry link. Implemented components, proposed solutions and future work are separated explicitly. No experimental performance of the complete system is reported; the design target is a waypoint error of 10–20 cm, to be validated in later stages.**

***Index Terms*—autonomous ground vehicle, differential drive, STM32, sensor fusion, odometry calibration**

I. Introduction

A. Background and Motivation

Autonomous ground vehicles (AGVs) integrate localization, perception, path planning and motion control so that a wheeled platform can perform a task without a human driver. In agriculture they are attractive because they reduce labor dependence and enable precise, repeatable operations; recent reviews name environmental perception, path planning and path-tracking control as the core technologies of autonomous field machinery [1]. Deployment remains difficult because of vegetation occlusion, uneven terrain, variable illumination and intermittent communications [2], and earlier surveys of agricultural off-road vehicles already identified Kalman filtering and machine vision among the key navigation tools [3]. Small educational platforms reproduce the same chain of problems (sensing, estimation, control and communication) at reduced scale.

B. The John Deere Challenge

The John Deere challenge of course MR2006B (Industrial Automation) requires each team to build a four-wheeled vehicle with differential or Ackermann steering [4]. Chassis, motors and drivers must be assembled by week 5; the vehicle needs at least one closed control loop that responds to a user reference, must send telemetry to an external system and must work in manual and automatic mode. In the final test it traverses a circuit on the paved paths of the Jardín de las Carreras (about 150 m, team estimate), avoids all grass, stops 10 s at each of three waypoints, returns to the start within 15 min, performs simple obstacle avoidance and runs on batteries. Grading rewards completing the circuit alone, using no external aids for the sensors, waypoint accuracy (2 points for 0–5 cm, decreasing logarithmically to 0 above 640 cm), obstacle avoidance (2 points for rerouting, 1 for waiting) and the exclusive use of an STM32 microcontroller [4].

C. Objectives and Engineering Requirements

Because speed is not graded, the objective is a vehicle that completes the circuit *reliably* rather than quickly. The main requirements are: (R1) four wheels, differential steering; (R2) at least one feedback loop responding to a user reference; (R3) manual and automatic modes; (R4) telemetry to an external system; (R5) circuit completion in under 15 min with 10 s stops, no contact with grass; (R6) obstacle stop and reroute; (R7) battery supply, STM32 as the only computer, no external aids. The team targets waypoint errors of 10–20 cm (1.5 of 2 accuracy points), since each halving of the error bracket yields only 0.25 points.

II. State of the Art

A. Autonomous Ground Vehicles and Agricultural Robotics

Reviews of agricultural navigation [1]–[3] describe a common architecture of perception, localization, planning and path-tracking control. High-precision positioning, typically GNSS combined with inertial sensing, is central in field machinery, while SLAM and vision are discussed for conditions in which satellite signals degrade under vegetation [1], [2]. Since the challenge excludes external aids and parts of the circuit lie under tree canopy, this design follows the relative-sensing branch: dead reckoning continuously corrected by the environment.

B. Ackermann and Differential Steering

In Ackermann steering the front wheels turn about separate axes so that all wheels roll about a common instantaneous center; the vehicle has a minimum turning radius and cannot rotate in place. In differential (skid) steering the left and right wheel speeds differ, the heading rate is $\omega = (v_R - v_L)/b$, and the vehicle pivots about its axle midpoint. Comparative studies of skid- and Ackermann-steered vehicles report different handling and energy trade-offs [5], [6]; a simulation study of an eight-wheeled vehicle found that skid steering consumed more energy than Ackermann steering in every maneuver except straight-line driving [6]. For narrow paths and ungraded speed, the ability to turn in place outweighs this penalty.

C. Sensors for Navigation and Obstacle Detection

Time-of-flight (ToF) sensors such as the VL53L1X measure up to 4 m at up to 50 Hz with a 27° typical field of view using a 940 nm Class 1 laser [7]; the usable range depends on ambient light, a key limitation outdoors. Wheel encoders provide the basic dead-reckoning signal, but systematic errors from unequal wheel diameters and uncertain track width accumulate without bound. The UMBmark procedure separates the two errors using clockwise and counter-clockwise square paths and reported an improvement of at least one order of magnitude in systematic odometry error [8]. Fusing encoders and an inertial measurement unit (IMU) in an extended Kalman filter (EKF) reduced position error by up to 47 % relative to unfiltered data in a recent differential-drive study [9].

D. Motor Control and Feedback

Brushed DC drives are commonly controlled by a feedforward term from the steady-state duty–speed model plus PI feedback. PI parameters for first-order-plus-delay plants follow from analytic rules such as SIMC [10], whose single tuning parameter is the closed-loop time constant. In cascaded loops, adjacent bandwidths are usually separated by a factor of 3–5.

E. Autonomous Navigation and Trajectory Following

Path tracking keeps a vehicle close to a reference path. Snider derived, tuned and compared several path-tracking methods for a car-like robot [11]; Giesbrecht *et al.* selected geometric pure pursuit for an unmanned ground vehicle for its accuracy, simplicity and robustness [12]; and a recent survey covers PID and pure pursuit for lateral and longitudinal control [13]. Pure pursuit is simple but tends to cut corners, so a linear cross-track/heading law with curvature feedforward is attractive when the path is known in advance.

F. Wireless Communication and Telemetry

Intermittent connectivity in the field [2] implies that autonomy must not depend on the link; telemetry is then used for monitoring and manual control. Low-cost serial radio modules such as the HC-12 provide a transparent UART bridge [14] but are half-duplex, so a slotted protocol with framing and checksums is needed to avoid collisions.

TABLE I
LITERATURE APPROACHES VERSUS THE TEAM'S DESIGN

Topic	Literature	This design
Positioning	GNSS + IMU; SLAM/vision [1], [2]	Encoders, gyro and ToF in an EKF; no GNSS (external aid)
Steering	Ackermann or skid [5], [6]	Differential, 2 driven wheels + 2 casters
Odometry	Calibration, e.g. UMBmark [8]	UMBmark planned; gyro supplies heading
Fusion	EKF, encoders + IMU [9]	5-state EKF with zero-velocity and boundary updates
Speed loop	Feedforward + PI, SIMC [10]	Implemented per wheel; gains from SIMC
Path tracking	Pure pursuit [11], [12]	Cross-track/heading law; pure pursuit as fallback
Telemetry	Serial radio [14]	HC-12 UART, framed slotted protocol

TABLE II
MAIN VEHICLE PARAMETERS

Item	Value (source)
Configuration	Differential; 2 driven wheels, 2 casters
Track width b	212 mm (geometric)
Drive wheel	∅76 mm nominal; 72.6 mm effective (calibrated)
Motor	JGB37-3530, 12 V, 178 rpm no load, 56.3:1
Encoder	Hall, 16 PPR/channel, ×4: 64 counts per motor rev.; 3595/3618 counts per wheel rev. (hand-measured)
Driver	2× BTS7960, 20 kHz PWM
Controller	NUCLEO-F446RE, 180 MHz, FPU
Battery	3S LiPo, 2.2 Ah, 35C
Vehicle mass	3 kg (assumed, not yet measured)
Cruise speed	0.3 m/s (design value)

G. Comparison with the Team's Design

Table I relates the approaches found in the literature to the present design.

III. System Design and Current Development

This section documents the work completed up to the previous challenge deliverable (state of October 2026). Measured values are taken from the team's technical reference [16].

A. Chassis and Mechanical Configuration

The chassis follows the differential-drive layout of Fig. 1: two driven wheels on a common axle through the body origin and two passive swivel casters that carry the remaining load. The concept envelope is 262×242 mm, so a turn in place sweeps a circle of 0.36 m diameter; the track width is $b = 212$ mm (Table II). The drive wheels are nominally ∅76 mm; a straight-line run calibrated the effective rolling diameter to 72.6 mm (the firmware reported 1000 mm while 955 mm was measured on the ground). The wheels sit on 6 mm D-shaft to 12 mm hex aluminum adapters, which keeps the final wheel size open until the ramp-lip height of the course is measured; caster size depends on the same measurement.

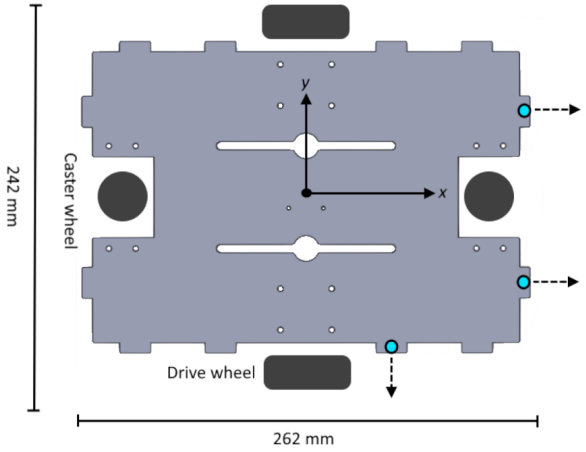


Fig. 1. Top view of the differential-drive layout, drawn from design parameters. Blue symbols: planned VL53L1X sensors (proposed, not yet installed).

B. Motor and Driver Selection

Brushed DC gearmotors with encoders were preferred over NEMA 17 stepper motors because an overloaded DC motor slows down and the encoder observes it, whereas a stepper loses steps silently. The selected JGB37-3530 has a catalogue torque of 3.4 kgf cm and a stall current of 4.3 A. A first-order torque check (assumed 3 kg, rolling coefficient 0.02, 8° grade, 0.3 m/s^2) gives about 0.10 N m per wheel, roughly 30 % of the catalogue value, so torque is not the limiting factor; traction and edge geometry are. Two BTS7960 half-bridge modules, one per motor, are driven at 20 kHz with module logic supplied at 3.3 V so that the 3.3 V STM32 outputs satisfy the input threshold [15]. An earlier opto-isolated L298-type driver was used for the first bench identification and was retired after a failure.

Bench tests without load confirmed the linear model $d = K_v n + K_s \operatorname{sgn} n$ ($R^2 > 0.999$, no-load speeds 181 and 177 rpm). Identification on the BTS7960 with the wheels in the air gave a first-order plant with static gain $K = 7.60$ (mm/s)/%, time constant $\tau_m = 50$ ms and a 7 % dead zone [16]. The two motors are not a matched pair (12 % speed and 20 % breakaway-duty difference at equal command), which the per-wheel feedforward and feedback must absorb.

C. Steering Mechanism

Steering is purely differential; there is no steering actuator. With wheel speeds $v_{L,R}$ and track width b,

$$v = \tfrac{1}{2}(v_R + v_L), \qquad \omega = \frac{v_R - v_L}{b}, \tag{1}$$

and the instantaneous center of rotation lies on the axle line, coinciding with the axle midpoint for a turn in place. Ackermann steering was rejected because it cannot rotate in place, needs a larger turning envelope at the ramp and earns no points in the grading scheme.

D. Material Selection and Engineering Justification

The selection criteria that follow from the requirements are: rigid mounting of the IMU at the axle midpoint (no elastic mounting), low mass for the assumed 3 kg vehicle, space for cabling without dangling wires, and procurement from local suppliers. For ease of manufacturing, it was determined that the vehicle chassis should be assembled from laser cut plates. The materials available for laser cutting were MDF (medium density fibreboard) and acrylic. To realize the aforementioned rigidity requirement, 6 mm thick MDF was selected as chassis material in spite of being more dense than acrylic. MDF also allows for easy mounting of electrical and mechanical components with bolt connections through drilled or laser cut holes and slots. The MDF parts are held together with friction but will eventually be glued together permanently.

TABLE III
COMPONENT STATUS

Component	Function	Status
NUCLEO-F446RE	Main controller	Implemented
Motors + Hall encoders	Traction, odometry	Implemented
2× BTS7960	Motor drivers	Implemented
Wheel cascade (50 Hz)	Speed/position loop	Implemented
HC-12 link, USB console	Commands, telemetry	Implemented
LiPo, buck, 3.3 V LDO	Power	Implemented*
IMU (6-axis)	Heading rate	Pending (part undecided)
8× VL53L1X	Edge, obstacles	Proposed
EKF	Pose estimate	Proposed
Path follower, mission logic	Automatic mode	Proposed
Framed telemetry protocol	Link robustness	Proposed
Course survey, UMBmark	Calibration	Future work

*Fuses, main switch and battery-voltage divider still to be fitted.

E. Current System Architecture

The STM32F446RE (Cortex-M4F, 180 MHz) executes all functions without an operating system. TIM1 generates the PWM, TIM3 and TIM8 decode the two quadrature encoders, TIM6 provides the 50 Hz control tick, USART1 serves the HC-12 radio and USART2 a USB console (115200 Bd). A 3S LiPo pack feeds a motor branch and, through a 5 V buck converter and a 3.3 V regulator, a logic branch; both join only at a star point. Table III lists each component, its function and its status, and Fig. 2 shows the architecture.

IV. Proposed Control and Automation Strategy

Everything in this section other than the wheel cascade is a proposal that has not yet been implemented or validated.

A. Manual and Automatic Operation

In manual mode the operator sends absolute set-points (v, ω) over the HC-12 link or the USB console; inverse kinematics from (1) gives the wheel speeds. A heartbeat failsafe will set $v = \omega = 0$ if no valid command arrives for 500 ms; a 2 s deadman timeout already protects open-loop console commands. In automatic mode the vehicle follows a pre-programmed path of LINE, ARC and TURN segments, each carrying its corridor and speed limit. A mission state machine sequences INIT (gyro bias convergence), DRIVE, APPROACH (0.1 m/s), HOLD (10 s) and TURN, with OBSTACLE_STOP, EVADE and FAULT as exceptional states. Autonomy does not depend on the radio link, but a received stop command is always honored.

B. Feedback Control Loop (Implemented)

Each wheel runs a cascade at 50 Hz in the TIM6 interrupt: a trapezoidal profile, a proportional position loop, and a speed loop with model feedforward and PI feedback,

$$u = \operatorname{sgn}(v_c)\big(K_s + K_v|v_c|\big) + K_v\tau_m a_p + K_p e_v + I_k, \quad (2)$$

where v_c is the commanded speed, a_p the profile acceleration, e_v the speed error and I_k the integrator clamped at $\pm 30\,\%$. A synchronization term couples the two wheels so that heading transients decay faster than distance errors. The gains follow the SIMC rules,

$$K_p = \frac{\tau_m}{K(\tau_c + \theta)}, \qquad T_i = \min\big(\tau_m,\, 4(\tau_c + \theta)\big), \quad (3)$$

with $\theta = 49$ ms (sample period plus speed filter) and $\tau_c = 2\theta$. This yields $K_p \approx 0.045\,\%/(\mathrm{mm/s})$, $T_i = 50$ ms, a design crossover of 6.8 rad/s and a phase margin of about 71°. These are design values computed from the identified plant, not measured closed-loop results. A 1000 mm test move ended at 999.5 and 1000.3 mm in the firmware (encoder) frame; independent ground measurements and floor tests under load remain to be done.

C. Sensor Integration

Heading is taken from a gyroscope rather than from the wheel difference, because the latter depends on the effective track width, which changes with slip. A five-state EKF with state $[x\ y\ \theta\ b_g\ s]^\top$ (position, heading, gyro bias, odometry scale) predicts from the encoder distance and gyro rate. Corrections are optional and gated by a Mahalanobis test: a zero-velocity update of the gyro bias during each 10 s stop, edge-crossing constraints from downward-looking ToF sensors, range and heading fits to straight faces from lateral sensors, and post landmarks for along-track position. The eight VL53L1X sensors have distinct roles: four look down (run validity, drop-off safety, EKF measurement, evasion bound), two look forward (reflex stop, free space) and two sideways. Having no fix is the normal case, so the circuit must be completable by dead reckoning alone.

D. Autonomous Trajectory Following

The reference is a geometric path decoupled from time. With cross-track error e_y, heading error e_ψ and path curvature κ, the path follower commands

$$\omega = \kappa v - k_\psi e_\psi - k_y e_y, \quad k_y = \frac{\omega_n^2}{v}, \quad k_\psi = 2\zeta\omega_n, \quad (4)$$

with $\omega_n \approx 1.5$–2 rad/s and $\zeta = 0.7$, which respects the bandwidth separation from the 6.8 rad/s speed loop; pure pursuit is kept as a fallback. Corners are driven stop–turn–go on level ground, since turning on a slope corrupts the integrated heading. Obstacle evasion is treated as a one-dimensional lateral offset search inside the measured corridor; the inner ground sensors remain armed as an interlock during evasion.

E. Data Transmission and Telemetry

Telemetry and remote control use the HC-12 on USART1 at 9600 Bd, already running in parallel with the USB console. Because 9600 Bd limits throughput, the proposed protocol uses COBS framing with CRC-16, absolute (idempotent) commands, and the vehicle as timing master, which avoids half-duplex collisions: a 24-byte binary frame needs 25 ms, i.e. 25 % of the channel at 10 Hz, against 62.5 % for a 60-character text line. The regulatory status of the 433 MHz band in Mexico and the use of radio modules with their own firmware under the "STM32 only" criterion are open questions for the course instructor.

F. Preliminary Block Diagram

Fig. 2 shows the relation between sensors, controller, drivers, actuators and communication. Solid green blocks are implemented; dashed blue blocks are proposed.

G. Expected Performance (Analytical, Not Validated)

An analytical error budget for a 40 m reference-free segment gives 21 cm lateral error for a $0.3°$ entry-heading error and 8 cm along-track error for a 0.2 % scale error after per-surface calibration [16]. Heading fits to straight faces and a landmark fix before each waypoint are expected to bring the error into the 10–20 cm bracket. These estimates are model-based and must be confirmed experimentally.

V. Future Work

Pending components. Select and order the IMU (the originally planned ICM-42688-P is unavailable; the LSM6DSV is the leading candidate); fit and test the eight VL53L1X sensors; fit fuses, main switch, battery-voltage divider and a hardware emergency stop; choose casters and wheel size after measuring the ramp lip; procure spares (Nucleo board, driver, motor); finalize the chassis and cabling.

Programming and integration. Move to a cyclic executive with a 1 MHz timestamp base; implement the EKF, path follower and mission state machine; add IMU (SPI with DMA) and two I^2C buses with per-sensor shutdown lines for address assignment; implement the framed telemetry protocol and the PC ground station; verify battery-only start-up (the clock currently derives from the ST-Link) and the thermal load of the 3.3 V regulator.

Validation. Survey the course (ramp lip and gradient, path widths, edging, landmarks); calibrate under load (K_v, K_s, per-surface scale); run UMBmark with a return-error target below 10 mm for a 2 m square [8]; characterize the gyro by Allan variance; test the EKF stationary and hand-pushed (10 m within 10 cm); close a 10×10 m rectangle within 30 cm; then complete the circuit with waypoints within 40 cm (week 15) and 20 cm (week 16), and log telemetry for 15 min at 10 Hz with fewer than 1 % lost frames. Open questions for the instructor concern the "four-wheeled" definition with two casters, the legal radio band, and the "STM32 only" rule for modules with their own firmware.

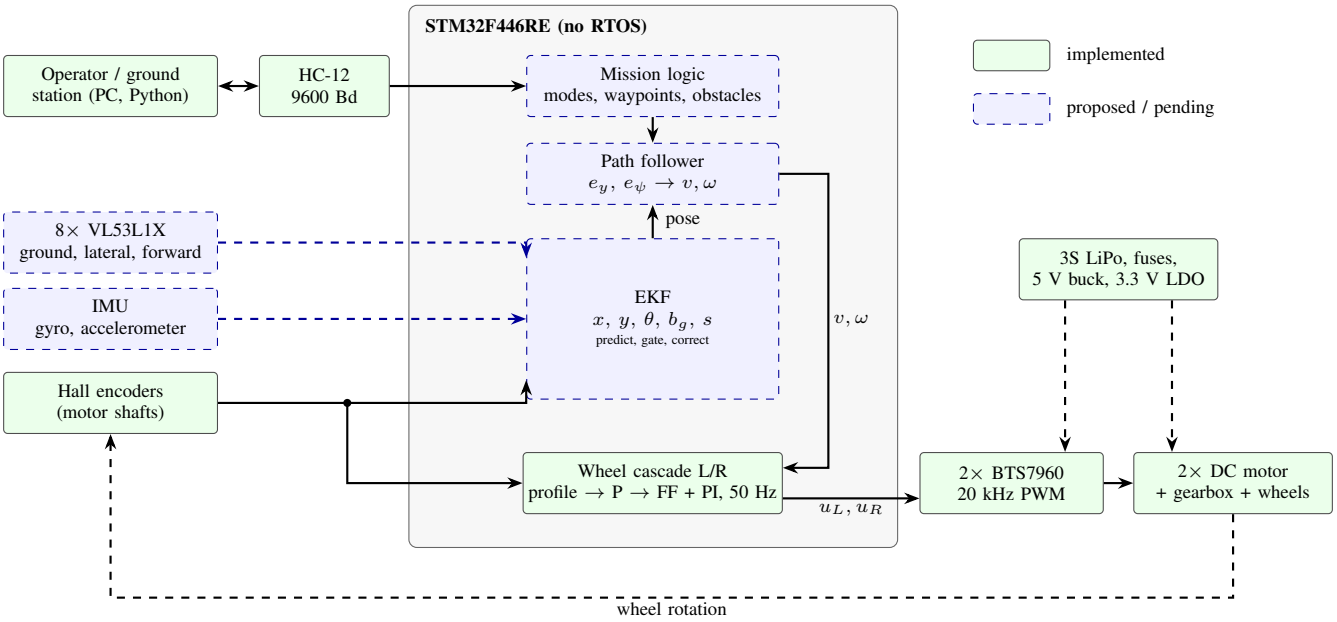


Fig. 2. Preliminary system block diagram: sensors, STM32 controller, drivers, actuators and telemetry link. The wheel cascade, drivers, motors, encoders and radio link exist; the estimator, path follower, mission logic, IMU and ToF sensors are proposed.

REFERENCES

[1] H. Wu, X. Wang, X. Chen, Y. Zhang and Y. Zhang, "Review on key technologies for autonomous navigation in field agricultural machinery," *Agriculture*, vol. 15, no. 12, Art. 1297, Jun. 2025.

[2] N. Boros, B. Ambrus and A. Nyéki, "Navigation and sensor fusion for autonomous field robots in precision agriculture: Narrative review," *Sensors*, vol. 26, 2026, doi: 10.3390/s26165169.

[3] H. Mousazadeh, "A technical review on navigation systems of agricultural autonomous off-road vehicles," *J. Terramechanics*, vol. 50, no. 3, pp. 211–232, Jun. 2013.

[4] Tecnológico de Monterrey, "Challenge MR2006B," course brief, Sensors & Actuators module, 2026.

[5] B. Maclaurin, "Comparing the steering performances of skid and Ackermann steered vehicles," *Proc. IMechE Part D: J. Automobile Eng.*, vol. 222, no. 5, pp. 739–756, 2008.

[6] J. D. Goodell, "Simulation and parameter sensitivity studies of skid-steering and Ackerman steering on an eight-wheeled vehicle," M.S. thesis, Univ. Texas at Austin, Austin, TX, USA, 2019.

[7] STMicroelectronics, "VL53L1X: A new generation, long distance ranging Time-of-Flight sensor based on ST's FlightSense technology," Datasheet DocID031281, Rev. 4, Oct. 2021.

[8] J. Borenstein and L. Feng, "Measurement and correction of systematic odometry errors in mobile robots," *IEEE Trans. Robot. Autom.*, vol. 12, no. 6, pp. 869–880, Dec. 1996.

[9] J. G. J. C. Amorim, M. F. Pinto, G. G. R. Castro, T. M. B. Santos, A. L. C. Canela and J. S. J. C. C. Amorim, "Optimized Kalman gain for enhanced localization in differential drive robots," in *Proc. SBR/LARS*, Vitória, Brazil, 2025, pp. 102–107.

[10] S. Skogestad, "Simple analytic rules for model reduction and PID controller tuning," *J. Process Control*, vol. 13, no. 4, pp. 291–309, 2003.

[11] J. M. Snider, "Automatic steering methods for autonomous automobile path tracking," Robotics Inst., Carnegie Mellon Univ., Pittsburgh, PA, USA, Tech. Rep. CMU-RI-TR-09-08, Feb. 2009.

[12] J. Giesbrecht, D. Mackay, J. Collier and S. Verret, "Path tracking for unmanned ground vehicle navigation: Implementation and adaptation of the pure pursuit algorithm," Defence R&D Canada – Suffield, Tech. Memo. TM 2005-224, Dec. 2005.

[13] H. Jain and P. Babel, "A comprehensive survey of PID and pure pursuit control algorithms for autonomous vehicle navigation," arXiv:2409.09848, Sep. 2024.

[14] HC Information Technology, "HC-12 wireless serial port communication module," User manual.

[15] Infineon Technologies, "BTS7960: High current PN half bridge," Datasheet.

[16] MR2006B team, "MR2006B vehicle technical reference," Rev. 1.0, internal document, Oct. 2026.